

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F
Q11	A	B	C	D	E	F
Q12	A	B	C	D	E	F
Q13	A	B	C	D	E	F
Q14	A	B	C	D	E	F

1. Which class of lever has the load situated in the middle? [1]

1st class lever

Reverse lever

Compound lever

4th class lever

3rd class lever

2nd class lever

2. How long must the designed hand hold an item to meet the stated testable goals? [1]

30 seconds

20 seconds

5 seconds

2 seconds

1 minute

10 seconds

3. Which term refers to how the anatomy of a real human hand influences the design? [1]

Societal Impact

Accessibility & Cost

Success Criteria Analysis

Critical Failures

Group Dynamics

Bio-mimicry Connection

4. Where is the effort located in a 3rd class lever? [1]

In the middle

Underneath the fulcrum

Mixed with the load

On the far left end only

Behind the pivot point

On the far right end only

5. What is the primary benefit of studying "bio-mimicry" (how the human hand works) before building the robot? [1]

To understand how nature creates movement and grip to solve engineering problems

To prove that robots are stronger than humans

To learn how to draw a hand for art class	Collaboration and teamwork
To see if human skin is waterproof	Advanced calculus
To count how many bones are in the wrist	Silent reading
To determine if the robot needs fingernails	Memorization of facts
	Handwriting speed
	Competitive arguing
6. Which anatomical component connects to the tendons and powers them? [1]	
The bones	9. Why is it valuable for engineers to design low-cost robotic hands instead of only expensive ones? [1]
The muscle	
The skin	To avoid using modern materials
The fulcrum	To make the hands heavier and harder to use
The wrist	To stop people from using robotic hands
The joints	To decrease the quality so the hand breaks faster
7. How is the "fulcrum" defined in relation to a lever? [1]	To make helpful technology accessible to people who cannot afford thousands of dollars
The weight of the load	To make sure only wealthy people can use them
The place where the lever or levers are free to rotate	
The place where you apply force	
The result of mechanical advantage	10. Why are robotic grippers necessary for industries like space exploration or handling hazardous waste? [1]
The rigid bar connecting the parts	Because robots look cooler than humans in photos
The object being lifted or moved	To use up extra electricity
8. What key skill are students practicing when they assign roles, share responsibilities, and resolve disagreements? [1]	Because robots get lonely in space
	To ensure the mission costs more money

To perform tasks that are too dangerous or deadly for humans to do directly

To prevent humans from having jobs

11. Why is the "Iteration & Redesign" step of the engineering process important for students? [1]

To make the project take as long as possible

To copy exactly what the teacher built

To use up all the remaining glue and tape

To learn from failures and improve the performance of the design

To make the robotic hand look more colorful

To prove that the first attempt is always perfect

12. What component of the human hand acts like a string to pull the fingers toward the palm? [1]

The fulcrum

The skin

The lever

The wrist

The tendons

The joints

13. Which part of the human hand provides enough friction so that gripped items do not slip away? [1]

The wrist bone

The bones

The knuckles

The joints

The skin

The muscles

14. A lever is used to give the user which specific benefit? [1]

Mechanical advantage

Increased friction

Thermal insulation

Reduced gravity

Electrical conductivity

A lighter load